

Parvez Akhtar

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PROFESSIONAL SUMMARY

Senior Applied AI Engineer with 5+ years building production-grade AI agents on top of high-scale distributed systems. Designs agentic workflows with multi-step orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK), tool/function calling over MCP (Model Context Protocol), and retrieval-based context management across large codebases and knowledge bases. Integrates and quantitatively evaluates multiple LLM providers (OpenAI, Anthropic, AWS Bedrock, Azure OpenAI) against task-specific benchmarks, working day-to-day across Python and TypeScript. Proven track record of taking experimental AI agents to reliable, cost-efficient production, reducing inference costs by ~90%.

TECHNICAL SKILLS

AI Agents & Orchestration: AI Agents & Multi-Agent Orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK, LangChain), Sub-Agents & Swarm Orchestration, Multi-Step Workflows, Tool/Function Calling, MCP (Model Context Protocol), Structured Outputs, Human-in-the-Loop, Agent Memory.

LLM Integration & Prompting: Multi-Provider LLM Integration (OpenAI GPT, Anthropic Claude, AWS Bedrock, Azure OpenAI), Prompt Engineering, Context Management (retrieval-based context, agent memory, semantic caching), Guardrails & Prompt-Injection Defense, RAG (Retrieval-Augmented Generation: Hybrid Search, Re-ranking), Fine-tuning (LoRA/QLoRA, Hugging Face), Embeddings, Vector DBs (Qdrant/PGVector).

LLM Evaluation & Observability: LLM Evaluation & Benchmarks (RAGAS faithfulness/answer-relevancy, CI/CD eval gates), Failure-Mode & Conversation Analysis, LLM Tracing (Langfuse, OpenTelemetry), SigNoz, ELK Stack, CloudWatch, Prometheus.

Languages: TypeScript, Python, Java, SQL.

Frameworks: FastAPI, Django, Spring Boot, PyTorch, TensorFlow, Scikit-learn, XGBoost.

Cloud & Infra: AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, Kubernetes, Docker, Terraform, Temporal.io.

Data Engineering: Snowflake Cortex, Apache Airflow, Kafka, Elasticsearch, ETL Pipelines, MongoDB, Redis, PostgreSQL, Browserbase.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Senior AI/ML Engineer, Lead Architect

Remote

04/2025 – Present

- As **Lead Architect**, drove cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous business problems into production AI-agent architectures.
- **Client, VTS (current), AI Document-Intelligence Agent:** Built an agentic **document-intelligence** pipeline that abstracts structured terms from commercial real-estate leases with **structured-output validation**. Diagnosed a **10-point accuracy regression** by analyzing **failure modes** across multi-turn extraction, then redesigned prompt strategy and image preprocessing to recover the full drop and restore field-level extraction to its **~88% baseline**, unblocking downstream automation.
- **Agentic Orchestration & Tool Use:** Orchestrated a serverless **multi-agent** framework for Salesforce automation with **tool/function calling** over **MCP (Model Context Protocol)** and **sub-agent (swarm) orchestration**. Cut idle compute via a pure **pay-per-execution** model (saving **~\$45k/year**).
- **Client, Outcomelabs.ai (GenAI Platform):** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **Strategy Router** that classifies each field by complexity and routes **~60%** to deterministic regex/rule-based parsers, reserving the LLM for the rest, cutting token spend **~40%** with no measurable accuracy loss.
- **Retrieval & Ranking:** Built a **Hybrid Search** pipeline for retrieval-based context, fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5 (Normalized Discounted Cumulative Gain)** by **~15 points (0.71 to 0.86)**.
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing fine-tuning strategies for cost efficiency.
- **Core Engineering:** Constructed an entity matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

Kuttl (Incubated at IIM Lucknow)

Remote

Founding Engineer

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the entire codebase and infrastructure, ensuring a **zero-downtime handover** of the platform to the incoming engineering team.

- Operationalized the LLM lifecycle with **CI/CD evaluation gates (RAGAS)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments that fell below threshold and catching quality regressions before production.
- **Fine-tuned Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction, quantitatively choosing the cheaper model after benchmarking, achieving **98% schema compliance** while cutting per-token inference cost by **90%** and latency by 300ms.
- Engineered a distributed document parsing pipeline using **FastAPI** and **Temporal.io** for multi-step workflow orchestration, processing 50k+ daily messages with resilient retry logic and dead-letter queues to ensure no data loss.
- Applied semantic caching and request batching for RAG (Retrieval-Augmented Generation) endpoints, improving **P99 latency by 40%** during high-concurrency bursts.
- Built the core backend and AI infrastructure from scratch, scaling the platform to handle peak loads while maintaining strict startup cost constraints.

Ampliflow

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.